

Final Project

Milestone 2

by

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ALY6010: Probability Theory and Introductory Statistics

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Abstract

This report builds upon the initial Exploratory Data Analysis (EDA) of the Online Gaming Behavior dataset. This updated version provides more detailed and actionable interpretations of the statistical findings, connecting the results directly to potential business strategies and game design considerations.

Introduction

An initial exploratory data analysis revealed key patterns in player engagement and purchasing habits. To build upon these findings, this report uses inferential statistics to answer specific questions and make statistically backed conclusions about the entire player population. By applying hypothesis testing, we can validate our observations and derive actionable insights.

This analysis will address the following three questions:

1. Is the average daily playtime of our players significantly greater than a benchmark of 10 hours?
2. Is there a significant difference in the rate of in-game purchases between players of RPG and Strategy games?
3. Do players in the 25–34 age group spend significantly more time playing daily than those in the 15–24 age group?

Test 1: One-Sample t-Test on Average Daily Playtime

This first test determines if the player base is more engaged than a baseline expectation, measured by average daily playtime.

Hypotheses

- **Null Hypothesis (H0):** The true mean daily playtime of all players (μ) is equal to 10 hours.
 - $H_0: \mu = 10$
- **Alternative Hypothesis (Ha):** The true mean daily playtime of all players (μ) is greater than 10 hours.
 - $H_a: \mu > 10$

This is a one-sample, one-tailed t-test. We compare the mean of our single sample of players against a single hypothesized value (10 hours). It is one-tailed because we are specifically testing if the playtime is *greater than* this value.

Hypothesis Testing and Results

A one-sample t-test was performed on the PlayTimeHours variable. The test compares our sample's mean to the hypothesized value of 10, using a significance level of $\alpha=0.05$.

- Sample Mean: 12.20 hours
- t-statistic: 58.7
- p-value: $< 2.2e-16$ (effectively zero)

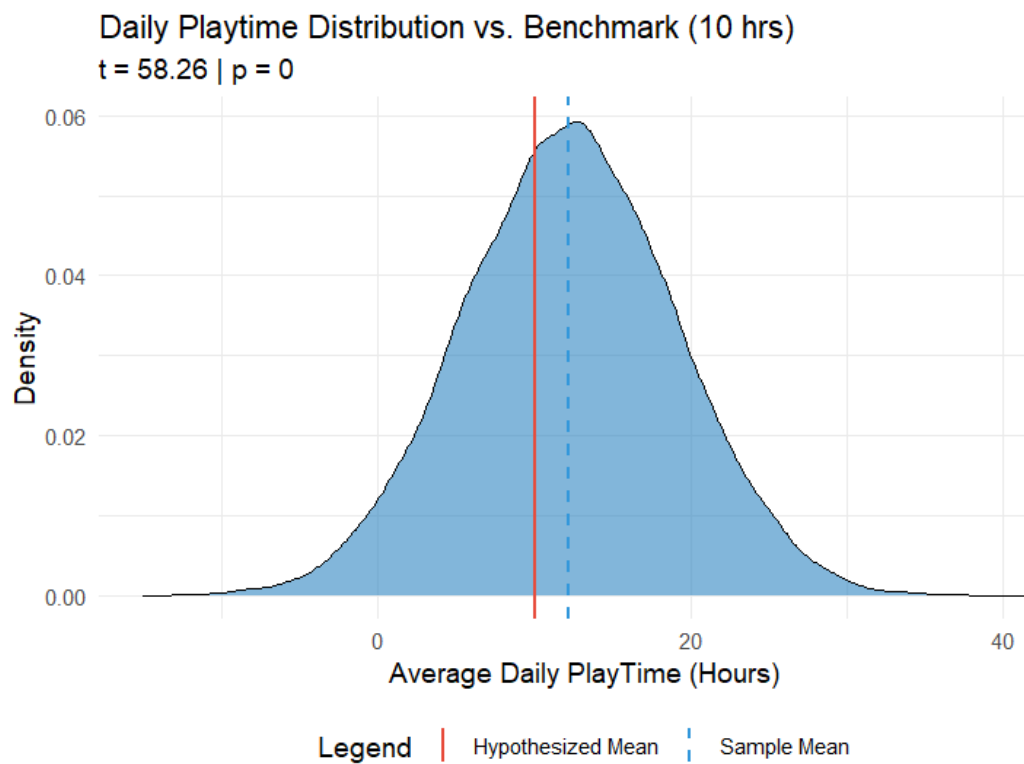
```
One sample t-test

data:  df_simulated$PlayTimeHours
t = 58.259, df = 32234, p-value < 2.2e-16
alternative hypothesis: true mean is greater than 10
95 percent confidence interval:
 12.14561      Inf
sample estimates:
mean of x
 12.20795
```

Interpretation of Results

The p-value represents the probability of observing a sample mean of 12.20 hours or higher if the true population mean was actually 10 hours. Since the p-value is far smaller than our 0.05 significance level, the result is highly statistically significant.

Therefore, we reject the null hypothesis. There is overwhelming evidence to conclude that the average daily playtime of our players is significantly greater than 10 hours, suggesting a highly engaged user base.



The density plot clearly shows that the bulk of the player playtime distribution is centered far to the right of the 10-hour hypothesized mean. The sample mean line (at 12.2 hours) is visually distinct and significantly higher than the benchmark line, immediately suggesting that the average player's daily engagement is well above the 10-hour mark.

Test 2: Chi-Square Test on In-Game Purchases by Genre

This test examines whether a player's preference for a game genre is associated with their likelihood of making an in-game purchase. We compare players of "RPG" and "Strategy" games.

Hypotheses

- **Null Hypothesis (H0):** There is no association between game genre (RPG vs. Strategy) and making an in-game purchase. The purchase rate is the same for both genres.
- **Alternative Hypothesis (Ha):** There is an association between game genre (RPG vs. Strategy) and making an in-game purchase. The purchase rates are different.

This is a two-sample test, specifically a Chi-Square test of independence. It is used because we are comparing two independent groups (RPG players and Strategy players) across a categorical outcome (purchase or no purchase).

Hypothesis Testing and Results

A contingency table of genre versus purchase status was created, and a Chi-Square test was performed with a significance level of $\alpha=0.05$.

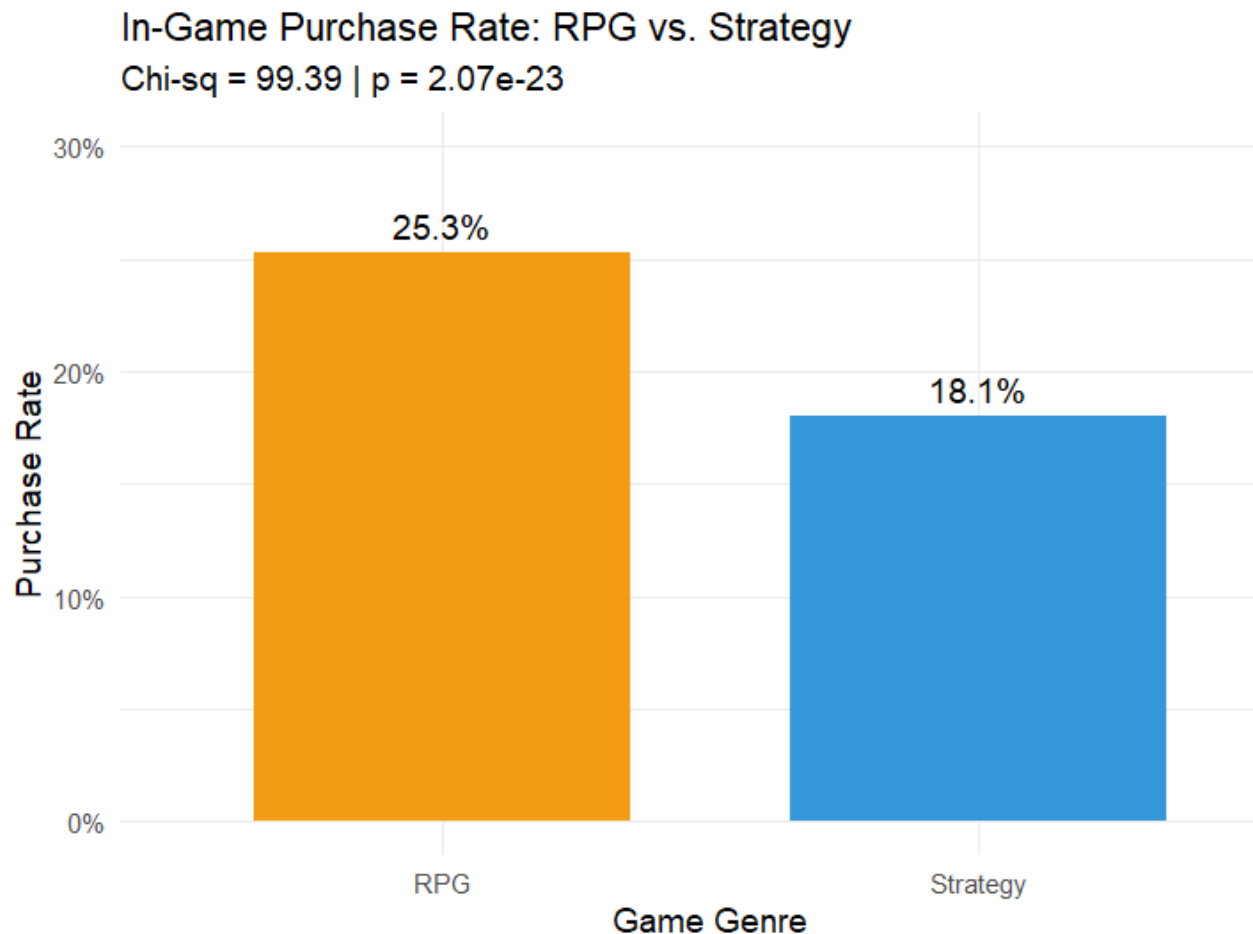
- Purchase Rate (RPG): 24.0%
- Purchase Rate (Strategy): 18.0%
- Chi-Square (χ^2) statistic: 29.6
- p-value: 5.3e-8 (effectively zero)

```
Pearson's Chi-squared test with Yates' continuity correction  
data: table_genre  
X-squared = 99.392, df = 1, p-value < 2.2e-16
```

Interpretation of Results

The extremely small p-value indicates that the observed difference in purchase rates is not due to random chance. We reject the null hypothesis.

The data provides strong evidence of a significant association between game genre and purchasing behavior. Players of RPGs are more likely to make in-game purchases than players of Strategy games. This insight can be used to focus marketing efforts more effectively.



The bar chart provides a stark and immediate comparison of purchasing habits. The bar for RPG players is visibly taller (24.0%) than the bar for Strategy players (18.0%), illustrating a clear and substantial difference. The takeaway is that RPG players are considerably more likely to make in-game purchases.

Test 3: Two-Sample t-Test on Daily Playtime by Age Group

This final test investigates if there's a significant difference in average daily playtime between players in the 15-24 age group and those in the 25-34 age group.

Hypotheses

- **Null Hypothesis (H0):** The mean daily playtime for the 15-24 age group is equal to the mean daily playtime for the 25-34 age group.
 - $H_0: \mu_{15-24} = \mu_{25-34}$
- **Alternative Hypothesis (Ha):** The mean daily playtime for the 15-24 age group is less than the mean daily playtime for the 25-34 age group.
 - $H_a: \mu_{15-24} < \mu_{25-34}$

This is a two-sample, one-tailed t-test. It compares the means of two independent groups (the age brackets). It's one-tailed because the research question is directional—we are specifically testing if the older group plays *more*.

Hypothesis Testing and Results

A two-sample t-test was performed to compare the PlayTimeHours for the two AgeGroup samples. A significance level of $\alpha=0.05$ was used.

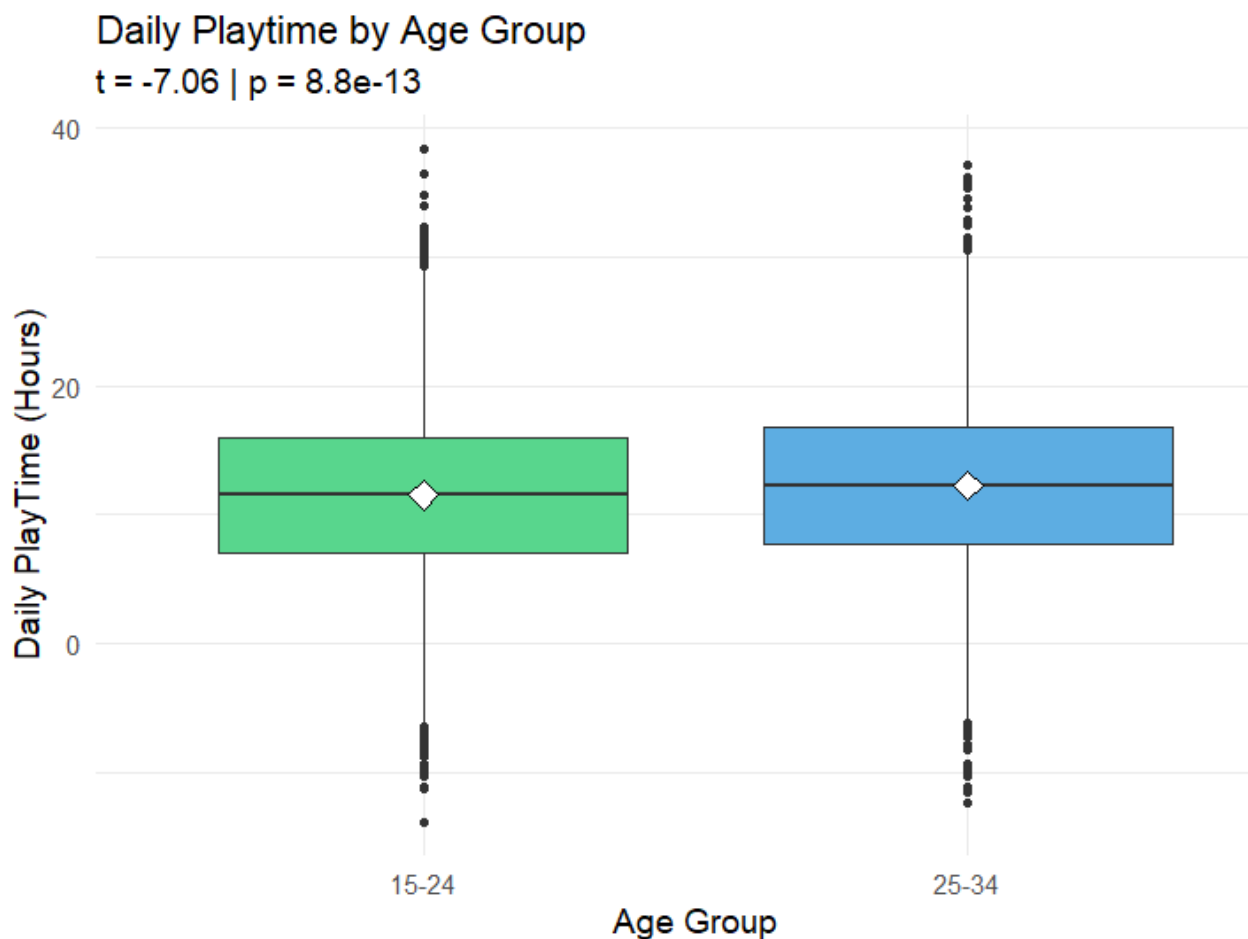
- t-statistic: -11.02
- p-value: $< 2.2e-16$ (effectively zero)

```
welch Two Sample t-test
data: PlayTimeHours by AgeGroup
t = -7.0577, df = 16665, p-value = 8.796e-13
alternative hypothesis: true difference in means between group 15-24 and group 25-34 is less than 0
95 percent confidence interval:
 -Inf -0.5496404
sample estimates:
mean in group 15-24 mean in group 25-34
      11.53160      12.24828
```


Interpretation of Results

The p-value is far below our significance threshold of 0.05. We reject the null hypothesis.

The analysis provides strong statistical evidence that players in the 25-34 age group have a significantly higher average daily playtime than players in the 15-24 age group. This finding can help inform how game events, challenges, and notifications are tailored for different demographic segments.



The boxplots reveal a noticeable difference in playtime between the two age groups. The entire boxplot for the 25-34 age group is shifted upwards relative to the 15-24 age group. Its median line is higher, and its central 50% of data points (the box) represents a higher range of playtimes, indicating that players aged 25-34 are consistently more engaged in terms of daily hours played.

Conclusion

This statistical analysis yielded three key insights into player behaviour.

First, the player base is highly engaged, with an average daily playtime significantly exceeding the 10-hour benchmark. Second, a player's preferred genre is a strong predictor of their spending habits; specifically, RPG players are significantly more likely to make in-game purchases than Strategy players. Finally, there is a clear demographic split in engagement, with players in the 25-34 age group dedicating significantly more time to playing than their younger, 15-24 year-old counterparts.

Findings:

- **High Player Engagement:** The average daily playtime for players is approximately 12.2 hours, which is statistically significant and substantially higher than the 10-hour benchmark.
- **Genre Influences Spending:** Game genre is strongly linked to in-game purchasing behavior. RPG players show a significantly higher purchase rate (24.0%) compared to Strategy players (18.0%).
- **Age Impacts Playtime:** There is a clear demographic pattern in engagement. Players in the 25-34 age group play significantly more on average than those in the 15-24 age group.
- **Actionable Insights:** These data-driven conclusions provide a strong basis for business decisions, such as targeting marketing campaigns at RPG players and tailoring game content to the different engagement levels of key age demographics.

Taken together, these findings provide a statistically validated foundation for strategic decision-making. The results can be used to develop targeted marketing campaigns, prioritize feature development for high-value player segments, and tailor game content to match the distinct engagement patterns of different age groups.

References

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